DATA ANALYTICS PROJECT

eCommerce Behavior Analysis

BUSINESS PROBLEM

Project Overview & Business Question

A multinational eCommerce company operating across Amazon, Jumia, and Souq wants to gain deeper insights into its sales performance, product demand, customer ratings, and revenue trends.

10,000

Dataset Rows

Transactional records across platforms with 11 key columns including customer demographics and purchase details.

Management has observed **fluctuations in monthly revenue**, varying product sales across platforms, and differences in customer satisfaction ratings among product categories – driving the need for structured data analysis.

Core Business Questions

Platform Revenue Performance

Which platforms – Amazon, Jumia, or Souq – generate the highest revenue and market contribution?

Customer Ratings & Sales Impact

How do customer satisfaction ratings correlate with sales performance and purchasing behavior?

Category Growth Drivers

Which product categories contribute most to overall business growth and revenue optimization?

Dataset Summary & Python EDA

Data preparation and exploratory analysis were conducted in Python, transforming raw transactional data into a clean, analysis-ready dataset loaded into PostgreSQL for downstream SQL querying.

Dataset Features

- **Rows:** 10,000 transactional records
- **Columns:** 11 key fields
- Customer: Order ID, City, Rating, Reviews
- Purchase: Product, Category, Price, Quantity, Order Date
- Platform identifiers across Amazon, Jumia, Souq

Python Preparation Steps

Data was systematically cleaned and enriched to ensure reliability and consistency before loading into the database for analysis.

- Loaded dataset with `pandas`; explored using `df.info()` and `df.describe()`
- Handled missing values and imputed nulls across key columns
- Renamed columns to snake_case for readability
- Removed redundant brand values (e.g. "Generic" vs "generic") and flagged ratings > 5
- Feature engineering: extracted `month`, `year`, and `day_name` from order date
- Connected to PostgreSQL and loaded the cleaned DataFrame for SQL analysis

Data Analysis Using SQL

Structured analysis was performed in **SQL to answer key business questions** across platforms, categories, products, and time periods – uncovering actionable patterns in sales and customer behavior.

Platform & Category Insights

- Revenue by platform – Souq, Jumia, Amazon compared
- Revenue by top-selling categories with quantities sold
- Revenue by cities ranked by total sales
- Average order value by category
- Platform revenue contribution percentages

Product & Rating Analysis

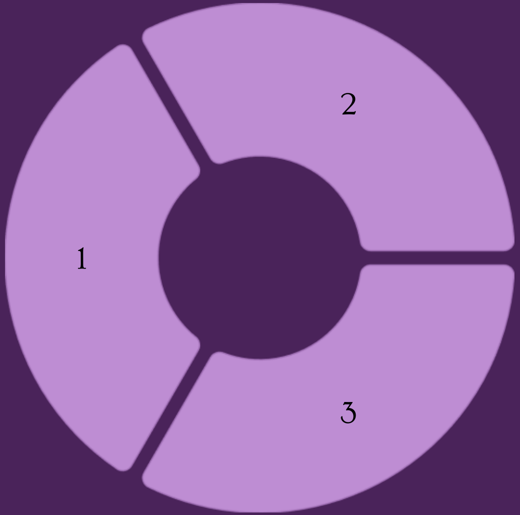
- Product performance by total rating and review count
- Sales and revenue compared across rating groups
- Top 5 brands by total revenue
- Top revenue-generating products by category and month

Time & Order Trends

- Highest sales month identified by total monthly revenue
- Orders by weekday compared across all days
- Weekday with the highest number of orders identified
- Monthly revenue trends to support inventory and marketing planning

VISUALIZATION

Power BI Dashboard



Revenue & Platform Metrics

Interactive visuals comparing platform revenue contributions across Amazon, Jumia, and Souq, with monthly trend lines to support strategic investment decisions.

Product & Category Performance

Category-level breakdowns of sales volume, average order value, and top-rated products — enabling stakeholders to spot high-growth and underperforming segments at a glance.

Customer Ratings & Behavior

Rating group comparisons linked to sales volumes, weekday order patterns, and brand-level insights to surface actionable customer satisfaction drivers.

DELIVERABLES

Project Deliverables

This project produces five structured deliverables that together provide a **complete, end-to-end data analytics solution** – from raw data cleaning through to strategic recommendations.

Data Preparation & Modeling

Python scripts to clean, transform, and engineer features from the raw dataset, including database integration with PostgreSQL.

SQL Data Analysis

Structured queries answering all key business questions on revenue, ratings, products, platforms, and time-based trends.

Also Included

- Power BI interactive dashboard with key visual insights
- Project report with findings and business recommendations
- Stakeholder presentation communicating actionable insights
- GitHub repository with all Python scripts, SQL queries, and dashboard files

Tech Stack

- **Python** – pandas for data loading, cleaning, and feature engineering
- **PostgreSQL / MySQL** – structured query analysis and business transactions
- **Power BI** – interactive dashboard for visual storytelling
- **GitHub** – version-controlled repository for all project assets

Business Recommendations

Based on the analysis of eCommerce sales and customer rating data, the following recommendations are designed to **optimize revenue growth, improve customer satisfaction, and strengthen platform strategy**.

🎯 Target Low-Performing Markets

Focus marketing campaigns on underperforming cities to improve customer reach and boost regional sales growth.

⭐ Promote Higher-Rated Products

Aggressively promote highly rated products, as customer ratings are directly linked to higher sales performance.

💰 Review Discount Policy

Balance sales boosts with margin control to ensure promotions and discounts drive profitable growth.

Strategic Priorities

Focus on improving low-performing categories and products through better pricing, promotions, and customer feedback analysis to raise overall product quality perception.

Platform Performance Focus

Leverage platform-wise revenue insights to invest more heavily in the best-performing markets like **Souq and Jumia**, while identifying growth opportunities on Amazon.

Product Positioning

Highlight top-rated and best-selling products within key brands to maximize profitability and strengthen brand equity across all platforms.

Analysis Methodology & Approach

Data Preparation

Clean and transform raw datasets

SQL Analysis

Query structured data for insights

Power BI Dashboard

Visualize trends and KPIs interactively

Recommendations

Deliver strategic findings to stakeholders

This end-to-end methodology ensures that raw transactional data from Amazon, Jumia, and Souq is systematically transformed into actionable business intelligence – enabling data-driven decisions across inventory planning, marketing investment, and platform strategy.